INCH-POUND

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MILITARY SPECIFICATION

WELDING SYSTEMS, ELECTRON BEAM, VACUUM TYPE, COMPUTER NUMERICAL CONTROLLED (CNC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers computer numerical controlled (CNC), electron beam welding systems of the vacuum type with equipment and accessories specified herein.

1.2 Classification. The welding systems shall be of the following classes and sizes. The class and size to be furnished shall be as specified (see 6.2).

Class I - Fixed Gun

Size 1	3.0 Kilowatt Beam Power
Size 2	6.0 Kilowatt Beam Power
Size 3	7.5 Kilowatt Beam Power
Size 4	15.0 Kilowatt Beam Power
Size 5	30.0 Kilowatt Beam Power
Size 6	42.0 Kilowatt Beam Power
Size 7	60.0 Kilowatt Beam Power

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3438

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Class II - Moveable Gun

Size 1 - 3.0 Kilowatt Beam Power

Size 2 - 6.0 Kilowatt Beam Power

Size 3 - 7.5 Kilowatt Beam Power

Size 4 - 15.0 Kilowatt Beam Power

Size 5 - 30.0 Kilowatt Beam Power

Size 6 - 42.0 Kilowatt Beam Power

Size 7 - 60.0 Kilowatt Beam Power

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified (see 6.2), the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications Forms Center, (ATTN: NPODS, 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

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2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

10 CFR 20 - Standards for Protection Against Radiation.

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AEROSPACE MATERIAL SPECIFICATION (AMS)

AMS 2681 - Electron Beam Welding, (DoD Adopted).

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-7511).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, and Tongues, (DOD adopted).

ANSI B46.1 - Surface Texture (surface roughness, waviness and lay), (DOD Adopted).

ANSI B93.114M - Pneumatic Fluid Power - Systems Standard for Industrial Equipment and Machine Tool Applications.

(Application for copies should be addressed to the American National Standards Institute ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

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AMERICAN WELDING SOCIETY (AWS)

AWS F2.1 - Recommended Safety Practices for Electron Beam Welding and Cutting, (DOD Adopted).

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th St., Miami, FL 33125-3136).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-227 - Tape perforated 1 inch, (DOD Adopted).

RS-244 - Character Code for Numerical Machine Tool Control Perforated Tape, (DOD adopted).

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machines, (DOD adopted).

RS-274 - Interchangeable Variable Block Data Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines, (DOD adopted).

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape, (DOD adopted).

(Application for copies should be addressed to the Electronic Industries Association, 2001 I Street, N.W. Washington, DC 20006-1899.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators, (DOD adopted).

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery, (DOD adopted).

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete system shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Design. The welding system shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The design shall include features and components necessary for accomplishing electron beam welding in a vacuum environment. The welding system shall have an electron beam gun mounted in or on a vacuum chamber. The vacuum chamber shall be designed to be evacuated to a desired level. Control of the system shall be directed by a computer numerical control (CNC) system. The CNC system shall control all machine operating components and systems, including linear and rotational axes, as specified. The machine and its numerical control system shall constitute a completely functional system with programmable operations that generate reproducible welding on a production basis. The welding system shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, and repair.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.4.1) or metric (see 6.4.2), as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate power (watts), amperage, voltage, and related electrical functions shall have independent zero adjustment and shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.2.2 Reclaimed materials. The welding system may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the welding system. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.2.3 Energy efficiency. The welding system and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All mechanical and pneumatic operating controls shall be located convenient to the operator's work station.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the welding system, whether or not specifically required herein, shall conform with all requirements of 10 CFR 20, 29 CFR 1910

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and AWS F2.1. If a conflict arises between 29 CFR 1910 and AWS F2.1, 29 CFR 1910 shall apply. Additionally safety and health requirements shall be as specified in detail (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial welding system shall be provided. The machine shall comply with all portions of AWS F2.1 relating to safety hazards.

3.2.5.1 Interlock switches. The welding system shall be provided with interlock switches for removing all power from the equipment when service doors that provide access to hazardous voltages are opened.

3.2.5.2 Safety Devices. Safety devices, safety interlocks, and fail safe features shall be provided to prevent the following:

a. Operation without proper cooling.

b. Operation of vacuum system at 10^{-3} mbar or less pressure without diffusion pump oil being at the proper temperature.

c. Exposure of hot diffusion pump oil to atmosphere.

d. Initiation of welding sequence before a vacuum is established.

e. Discharge of particulate matter into the atmosphere.

3.2.6 Mercury restriction. The welding system shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the welding system.

3.2.8 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the welding system shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation .

3.2.9 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The welding system shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the welding system to its maximum rating and capacity without permanent distortion or failure.

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3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or paste shall not be used for reclaiming any defective parts. Such processes may be used only for enhancing surface finish and appearance. Any castings employed which will be exposed to a vacuum environment shall be vacuum impregnated.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the welding system shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the welding system and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will transmit the full rated torque of the drive motor without damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem class and the emission and susceptibility requirement shall be as specified.

3.4 Components. The welding system shall consist of not less than a vacuum chamber, vacuum pumping system, electron gun, power supply, computer numerical control system, control station, pneumatic system, coolant system, and electrical system.

3.4.1 Vacuum chamber. The vacuum chamber shall be capable of withstanding internal operating pressures of 1×10^{-6} mbar. At a working pressure of 1×10^{-4} mbar, no in-leakage shall be detectable at a density of 2×10^{-8} standard cc per second of helium. Starting at a pressure of 1.3×10^{-3} mbar over a 4 hour period, the average pressure rise shall be no more than 7×10^{-3} mbar per hour for 10,000 liter and greater chambers, 13×10^{-3} mbar per hour for 1000 to 10,000 liter chambers, and 20×10^{-3} mbar per

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hour for 1000 liter and less size chambers. Surface texture of vacuum exposed surfaces shall have a finish of not greater than 32 roughness average (RA), in accordance with ANSI B46.1. The chamber size shall be as specified (see 6.2).

3.4.1.1 Chamber door. The vacuum chamber shall be furnished with a door which shall provide full access clearance equal in cross section to the inside dimensions of the chamber. A chamber door of sufficient size for an operator to physically enter shall require a pinning or locking assembly. A door of this size shall be designed for easy entry and exit, emergency exit, safe operation, and prevent improper access to the welding chamber. The inside surface of the door shall be not greater than 32 RA in accordance with ANSI B46.1.

3.4.1.2 Viewing ports. The vacuum chamber shall contain at least one viewing port. Any chamber viewing port provided shall be constructed of glass specifically selected for both its vacuum and radiation integrity. The chamber viewing port shall include an easily removable glass or plastic inside cover that acts as a disposable, metallic vapor deposition shield.

3.4.1.3 Feed throughs. The vacuum chamber shall be equipped with all necessary interconnecting feed-throughs for electrical power, coolants, gauges, back-fill gas, controls, and at least one spare multi-pin electrical connector.

3.4.1.4 Chamber lighting. Suitable lighting fixtures shall be provided for full view, non-glare illumination of the work area. Each light shall be not less than 100 watts, operate at no more than 125 volts, 60 Hz and be connected to the power source with a three wire grounded cable.

3.4.1.5 Pressure equalizing valve. A pressure equalizing gas valve for admitting air at atmospheric pressure to equalize chamber pressure within three minutes or less shall be furnished. This valve shall be equipped with a filter and shall suppress noise. When specified (see 6.2) chamber venting shall be accomplished with pressurized bottle gas.

3.4.1.6 Gun de-energizer. Means shall be provided for de-energizing the electron beam gun whenever the chamber is accessed.

3.4.1.7 Driving mechanism. The interior of the chamber shall be arranged to mate with and accept external and internal driving mechanisms for internally mounted fixturing.

3.4.1.8 Feed rates. Feed rates shall be externally controlled and adjustable over the nominal range from 5 to 100 inches per minute. Limit switches shall be provided to prevent overdriving motion devices. A method for reading travel speed directly shall be provided.

3.4.2 Vacuum pumping system. The vacuum pumping system shall have the capacity to evacuate and maintain a vacuum level of 8×10^{-5} mbar or greater in a clean, dry, and empty vacuum chamber, unless otherwise specified (see 6.2). The system shall be capable of evacuating the chamber to a working

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pressure of 1 X 10^{-4} mbar or less within a time period to be specified in 6.2. Electrical power shall be supplied to the vacuum system so that other equipment may be shut down without interrupting the power supplied to the vacuum equipment. Mechanical pumps shall be located to minimize or eliminate the transmission of vibration to the rest of the machine.

3.4.2.1 Pressure sensing instrumentation. The vacuum system pressure sensing instrumentation shall be of the manufacturers current design, capable of monitoring vacuum levels at critical locations and providing this information to both the operator and appropriate switching devices of the machine.

3.4.2.2 Vacuum sequencing. The system pumping operation shall be sequenced automatically or by a single sequencing switching device to evacuate the chamber to the proper welding vacuum level. The vacuum and welding sequencing controls shall be interlocked so that the electron beam cannot be initiated or maintained unless the pumping station has evacuated the chamber to the proper pressure.

3.4.3 Electron gun. The electron gun shall be either fixed (class I) or moveable (class II) as specified (see 1.2). The electron beam gun shall be designed to be operated in a continuous fashion and to maintain a constant beam centerline axis, over the full range of rated beam power output. The Class I machine's (fixed gun) electron beam gun and its associated components shall be enclosed in a vacuum tight column capable of sustaining a 1 X 10^{-5} mbar or lower pressure. This column vacuum shall be independent of chamber pressure. The Class II machine (moveable gun) shall incorporate this feature only when specified (see 6.2). A quick change filament shall be included in the electron gun and shall be easily and quickly replaceable in accordance with the manufacturer's standard commercial practice. The gun shall be designed to facilitate routine maintenance and cleaning. The gun shall be controlled by the CNC.

3.4.3.1 Beam alignment deflection. When specified (see 6.2), the beam shall be aligned by deflection to a minimum of two degrees to all sides of the beam centerline axis.

3.4.3.2 Beam manipulation deflection. The beam shall be manipulated by deflection to a minimum of five degrees to all sides of the beam centerline axis.

3.4.4 Power supply. All power supplies required for operation of the welding system shall be furnished. All power supplies shall be designed for continuous duty operation at rated load voltage and current. Voltage stabilization shall be $\pm$ 1% for a 10 percent variation in line voltage. Ripple effect shall not exceed 2 percent RMS with an output stability of $\pm$ 1 percent over a four hour period.

3.4.5 Computer numerical control system (CNC). The control system shall be the solid state, soft wired, computer numerically controlled (CNC) type. The system shall recognize EIA RS-274 input format. The system shall be of microprocessor based hardware architecture or a minicomputer to suppport a multi-programming/multi-tasking operating system specifically designed for

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application to CNC. The system shall execute part program data from CNC memory loaded from the input media specified herein. Plug-in circuit boards shall be coded to facilitate identification and location in the control and shall be arranged as convenient plug-in elements of quick disconnect modules with minimum connectors. Built in test points, test procedures, and instructions shall be provided for isolation and identification of defective components and circuitry. The microprocessor system shall be provided with a minimum of 256k bytes of user accessible main memory. Microprocessor main memory shall be expandable in increments of not less than 256k bytes. The minicomputer system shall have a minimum of 64k bytes for part programs and 64k bytes for executive software for a total of 128k memory. The system shall have off line storage capability. The control memory shall be non-volatile or have battery backup to prevent loss of program memory for not less than 72 hours. In the event of power loss, crossover from line to alternate power shall be automatic and shall signal the system to begin an orderly shutdown of processing. The system shall bring itself down in such a manner as to protect the machine and itself from damage or loss of data. Recovery from power outages shall be automatically or manually initiated upon power resumption and shall reload or reboot the executive program and bring itself back to a state where the operator can initiate part program load and restart. The system shall accept and translate program data into positioning and feed motion command signals for all axes drives. Speeds, direction, and path of axes movements, electron gun power, and sequence and timing of all operations shall be directed by the control. Axes and motion nomenclature shall conform to EIA RS-267. A line voltage of ±10 percent from normal shall not adversely affect the numerical control function. The system shall function in ambient temperatures ranging from 40° F to 95° F and relative humidities ranging from 5 percent to 95 percent non-condensing. The system shall automatically shut down when internal operating temperature exceeds safe operating temperature. System resolution shall be 0.001 inch for each linear axis and not more than 0.01 of a degree for each rotary axis. The axes shall be identified in accordance with EIA Standard RS-267. The system shall initiate an error or halt signal should a fault condition occur in the system.

3.4.5.1 Data input.

3.4.5.1.1 Punched tape input. When specified (see 6.2), the input media shall be a 1-inch, 8-channel punched tape. The input format and codes shall comply with EIA standards RS-227, RS-274, and RS-358. When specified (see 6.2), RS-244 shall be used in lieu of RS-358. The control shall be equipped with a photoelectric tape reader capable of holding, and reading tape prepared in the format specified at a rate of not less than 150 characters per second. Unless otherwise specified (see 6.2), the reader shall be supplied with standard take-up spools 5-1/4 inch in diameter. The tape reader shall be housed in a dust free environment. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA RS-244.

b. Even parity per EIA RS-358.

c. Misaligned sprocket holes.

d. No sprocket holes.

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Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

3.4.5.1.2 Hardshell diskette input. When specified (see 6.2), the input media shall be a 3-1/2 inch hardshell diskette with a minimum capacity of 720k bytes. The disk shall be double sided and double density. The input device shall be a diskette reader/writer of either the peripheral or integrated type as specified (see 6.2).

3.4.5.1.3 Manual data input (MDI). A means for manual data input MDI shall be furnished to permit manual entry of alphabetical, numerical, and special characters necessary for control of all machine functions. The manual input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.5.2 Data output.

3.4.5.2.1 Tape punch unit. When specified (see 6.2), machines shall have tape punch units for producing punched tape from edited programs of tape input and from programs stored in memory. The unit shall have a receptacle and circuitry that is compatible with the tape punch output interface required above for receiving signals and punching both paper and mylar tapes of dimensional characteristics, and coded in accordance with paragraph 3.4.5.1. The punching speed of the unit shall be not less than 75 characters per second.

3.4.5.2.2 Line printer. When specified (see 6.2) a printer capable of printing not less than 30 characters per second shall be provided. It shall be capable of printing not less than 132 columns of data. The printer shall provide data as an output from the computer in the form of typed copy.

3.4.5.2.3 Hardshell diskette output. When specified (see 6.2), the output media shall be a 3-1/2 inch hardshell diskette with a minimum capacity of 720K bytes. The disk shall be double sided and double density. The output device shall be a diskette reader/writer of either the peripheral or integrated type as specified (see 6.2).

3.4.5.3 Standard Control features.

3.4.5.3.1 Departure control. The maximum departure permitted by a single command in either incremental or absolute mode shall be not less than plus or minus 32.767 inches. The control shall automatically control the acceleration and deceleration of the workpiece. The resolution of workpiece movement in either direction shall be not more than 0.001 inch.

3.4.5.3.2 Absolute/incremental control. The control shall have absolute/ incremental input capability.

3.4.5.3.3 Part program edit. A program edit capability shall be provided to allow block insertion, deletion, or modification of any part program in the part program memory. Data related to part program edit shall be entered by the

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manual data input provided on the control panel. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.5.3.4 Feed rate override. When specified (see 6.2), a feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from zero to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that programmed.

3.4.5.3.5 Block delete. The control shall have a block delete feature to permit by-passing blocks of programmed data.

3.4.5.3.6 Buffer Storage. The master control unit (MCU) shall have buffer storage capable of storing one or more blocks of program data, read from the type of input specified. As blocks are transferred to active command, new blocks shall read into the buffer.

3.4.5.3.7 Block-by-block read. The control shall have a block-by-block read function to provide for reading information one block at a time.

3.4.5.3.8 Automatic cycle. The control shall have the capability of running in automatic cycle. Once programmed, the control shall automatically run until either a PROGRAM STOP or END-OF-PROGRAM command is encountered.

3.4.5.3.9 Fixture compensation. A fixture compensation feature shall be provided to permit compensation for fixture position errors.

3.4.5.3.10 Reversal error compensation. Electronic compensation within the control unit shall be provided to reduce the effects of reversal error. Means shall be provided to adjust the anti-backlash device and to input new compensation values, if applicable, to correct for wear and usage during the life of the machine.

3.4.5.3.11 Reference zero. The control unit shall have a reference zero to move each axis to a reference point to establish synchronization and reset all registers to zero.

3.4.5.3.12 Set zero. The control unit shall have a "set zero" feature to permit the specified axis register to be reset to zero establishing present axis location as zero.

3.4.5.3.13 Diagnostic capability. The control unit shall include a diagnostic system for monitoring the control during operation and for locating malfunctions when troubleshooting. It shall bring the system to a controlled stop if a malfunction occurs during operation. The system shall give fault data in understandable code or message form on the display unit which will identify a problem at board level in the controls and identify program errors.

3.4.5.3.14 Error compensation. Error compensation shall be provided to correct repeatable errors in axis calibration due to inaccuracies. A means shall be provided to input new error compensation values as required.

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3.4.5.3.15 Control features. The control unit shall display on the CRT the active line and a minimum of the next two lines of information in both single block and automatic modes. The screen shall be automatically updated. The control shall be capable of accepting subroutines that are recallable an unlimited number of times and capable of providing file management of more than one part program. The control shall be capable of entering a block of data into memory upon command. The control shall be capable of accepting program data in one type measurement system (either inch-pound or metric) as specified by the procuring activity (see 6.2). Minimum controls for operator displays shall be:

a. Mode of operation (i.e. automatic, single block, and manual data input (MDI).

b. Machine status (i.e. power on, in-cycle, idle, end of cycle, in-hold, and in E-Stop).

c. Control status (i.e. manual override, manual feedrate override, in-edit, and program transfer active).

d. Machine position.

e. Programmed position.

f. Program listing.

g. Current active block.

h. Next block.

i. Not used.

j. Operator warning, information messages, and diagnostics.

k. Active axes home position values.

3.4.5.3.16 Additional features. Unless otherwise specified (see 6.2), the following additional CNC features shall be provided:

a. Incremental and continuous jog for each axis.
b. Leading zero suppression or decimal point programming.
c. Means to restrict access to those personnel authorized.

3.4.5.3.17 Heat treat control. When specified (see 6.2), the control shall have the capability to direct the welder thru reproducible programmed operations necessary to heat treat on a production basis.

3.4.5.3.18 Program tryout. The control shall have a program tryout feature to permit new part programs to be tested through the control without machine motions.

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3.4.6 Control station. All control elements required for system setup and run operation shall be contained in an operator's control station. This station shall be located at a position convenient for normal system operations. All keys, dials, and switches shall be functionally grouped.

3.4.6.1 Emergency stop button. A mushroom head, red, "Emergency Stop" button shall be provided on the control station of all machines. An emergency stop button or other safety device shall be provided inside the vacuum chamber and on all control pendants when the vacuum chamber is of sufficient size for the operator to physically enter.

3.4.6.2 Start/stop buttons. Functional sequence "start/stop buttons" shall be provided on the control station.

3.4.6.3 Vacuum indicators. Vacuum indicators shall be provided on the control station.

3.4.6.4 Display. Unless otherwise specified (see 6.2), the display shall be a cathode ray tube (CRT). The CRT shall be capable of displaying machine system parameters and program codes from memory, buffer storage, and active data. The CRT shall be capable of displaying 256 characters or more.

3.4.7 Tracking system. When specified (see 6.2), a tracking system shall be provided for automatically maintaining the alignment of the electron beam with the weld joint. Not less than 0.010 inch (0.25 mm) of parent metal from each side of the joint shall be included in the tracking scan. The system shall include appropriate controls. The data acquired during the joint tracking shall be inserted in the computer memory for subsequent use.

3.4.8 Optical viewing system. When specified (see 6.2), an optical viewing system shall be provided and shall permit magnified viewing of the workpiece. Magnification shall be at least 5 power. The system shall assure an adequate viewing of the beam and the workpiece and shall incorporate an optical reticle with graduations. The optical system shall have provisions for introducing clean optical surfaces after welding degrades existing optical surfaces. An infinitely variable polarized filter or iris shall be provided. The viewing system furnished shall include all necessary components to assure in line adjustment between beam centerline and optical centerline.

3.4.9 Work handling systems.

3.4.9.1 Universal rotary tables. When specified (see 6.2) a universal rotary table shall be furnished. It shall be used to position workpieces about an axis which can be varied from a vertical to a horizontal plane. The table shall mount on the machine carriage mechanism. The angular motion of the table shall be motorized and designed for operating in the vacuum environment. The table spindle support structure shall pivot about a fixed axis provided by the integral platform base. The spindle shall be powered by the accessory drive system or by other means adaptable to the vacuum atmosphere. The table fixture plate shall be precision machined and shall provide piloting features and four radial "T" slots, located at 90 degrees to

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each other, for work fixture attachment. The "T" slots shall conform to ANSI B5.1. The fixture plate shall be constructed of stainless steel. Eccentricity of the spindle in operation shall be not greater than 0.001 inch total indicator reading (TIR) for tables of diameters up to 60 inches and 0.002 inch for tables of diameters above 60 inches. Speed regulation shall be compatible with the welding characteristics of the machine and shall be within two percent of set values. The table shall be capable of positioning automatically at 0.010 degree with variable feed rates to any of the 360,000 positions. The table tilt for "A" axis capability shall be not less than 90 degrees. Positioning and repeatability accuracies shall conform to table I.

TABLE I. Accuracy Tolerances of CNC.

"X" & "Y" axes positioning and repeatability accuracy:	
absolute positioning	±0.0015 in per ft
position repeatability	±0.001 in
"Z" axis positioning and repeatability accuracy:	
absolute positioning	±0.0015 in per ft
position repeatability	±0.001 in
"A" axis (when furnished) positioning and repeatability accuracy:	
absolute positioning	±0.01 degree
position repeatability	±0.01 degree

3.4.9.2 Combination linear rotary table. When specified (see 6.2), a combination linear rotary table shall be furnished. It shall be used to rotate or position a workpiece about the vertical axis of travel or position the workpiece along the horizontal axis. The combination table shall provide movements for producing a range of circular and girth welds around the vertical axis, linear welds along the vertical axis and linear welds along the horizontal axis. The table top shall be mounted to a precision bearing mounted carriage and power driven through a suitable drive arrangement. Means shall be provided for producing angular welds, such as positioning the gun at the desired angle on an adjustable mounting surface. The fixture plate shall be constructed of nickel plated mild steel or aluminum, and shall be provided with four radial "T" slots located at 90 degrees to each other. The "T" slots shall conform to ANSI B5.1. The plate shall have a center pilot hole of sufficient diameter to facilitate work fixturing. Eccentricity at the periphery of the table at any speed as stated below, shall be not greater than 0.005 inch for a table up to 60 inches X 60 inches and 0.010 inches for tables above 60 inches X 60 inches. Flatness of the fixture plate shall be within 0.005 inch TIR For a table up to 60 inches X 60 inches and 0.010 inches TIR for tables above 60 inches X 60 inches. Speed regulation of the table shall be compatible with the welding characteristics of the machine, and speed regulations rating shall be within two percent of set values. Positioning and repeatability accuracy shall conform to table I.

3.4.9.3 Horizontal rotary table. When specified (see 6.2), a horizontal rotary table shall be furnished. It shall position and rotate a workpiece about its vertical axis. The table shall be mounted on precision bearings and

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shall be power driven through a suitable gearing arrangement. The fixture plate shall be constructed of aluminum or nickel plated mild steel and shall be provided with four radial "T" slots located at 90 degrees to each other. The "T" slots shall conform to ANSI B5.1. The plate shall have a pilot hole to facilitate work fixturing. Eccentricity of the spindle at the operational speed shall be not greater than 0.002 inch TIR for table diameters up to 60 inches and 0.004 inches for tables above 60 inches diameter. Flatness of the fixture plate shall be within 0.005 inch TIR for tables up to 60 inches diameter and 0.010 inches for tables above 60 inches diameter. Speed regulation of the table shall be compatible with the welding characteristics of the machine, and speeds regulation rating shall be within two percent of set values. Positioning and repeatability accuracy shall conform to table I.

3.4.9.4 Headstock and tailstock. When specified, (see 6.2), a headstock and tailstock shall be furnished. The headstock shall be compatible with the tailstock and controlled by the CNC. The headstock and tailstock shall be equipped with a faceplate and the plate shall be provided with four radial "T" slots at 90 degrees to each other for work fixturing. The "T" slots shall conform to ANSI B5.1. The assembly shall be designed to mount on the work carriage. The major structure shall be fabricated of nickel-plated mild steel, with the faceplate made of stainless steel. The axis of rotation of the headstock shall be the same height as that of the tailstock. Spindle nose mounting surfaces for faceplates up to 60 inches in diameter, at maximum speed, shall have a runout within 0.002 inch TIR eccentricity and 0.004 inch above 60" diameter. The flatness of each faceplate shall be within 0.005 inch TIR. For plates up to 60 inches in diameter and 0.010 inch for plates above 60 inches diameter. The headstock shall be power driven through a suitable driving arrangement. Speed regulation shall be compatible with the welding characteristics of the machine. Speed regulation rating shall be within two percent of set values. The maximum length, maximum part diameter, and speed range shall be compatible with the capabilities of the welding machine. The tailstock spindle shall mount in precision ball or roller bearings. The axis of rotation of the tailstock shall be machined to the same height as the headstock or, when furnished, the universal rotary table. The spindle shall have no more than 0.002 inch TIR eccentricity on the pilot diameter.

3.4.9.5 Eccentric table. When specified (see 6.2), an eccentric table shall be furnished. The table shall be controlled by the CNC. The eccentric table shall mount directly on the machine work table. The work mounting surface of the eccentric table shall be a flat rectangular plane having tapped holes for fixtures. The table shall be driven either by an accessory drive package or by an integral drive motor. The top of the eccentric table shall be supported on ball bearings and guide bars which allow motion along either horizontal axis. The table top shall be driven by an adjustable eccentric head and the motion derived from this eccentric head shall combine the "X-Y" motions to produce a true circular motion of the workpiece. The eccentric head and the motion derived from this eccentric head shall combine the "X-Y" motions to produce a true circular motion of the workpiece. The major structures shall be fabricated from nickel-plated mild steel. The eccentric head shall be adjustable to provide the desired circle diameter for the workpiece. As the eccentric head is rotated, it shall drive the table along a

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coordinated "X-Y" path to trace out the required circle. Indexing of the part under the beam impact point prior to welding shall be accomplished by use of the "X-Y" machine motions independently of the eccentric table. Welding of any other circles of the same diameter within the area of the work mounting surface shall be possible by indexing the seam under the beam impact position and initiating eccentric table motion. The speed range of the table shall be adjustable from not more than 0.25 to not less than 23 RPM. The top surface of the eccentric table shall be flat within 0.005 inch TIR. The eccentric table shall have a speed regulation rating within two percent of set values. Positioning and repeatability accuracies shall conform to table I.

3.4.10 Run-out platform. When specified (see 6.2), a run out platform shall be furnished. The run-out platform, when installed, shall facilitate moving the workpiece into the work space. The run-out platform shall consist of a set of hardened precision ways which shall be constructed so as to clear the opening and closing of the chamber door. The ways shall be of corrosion resistant material or finish. The run-out platform shall be firmly anchored to the machine base or furnished with a moveable die table.

3.4.11 Accessory drive unit. When specified (see 6.2), an accessory drive unit shall be furnished. The accessory drive unit shall provide a rotary motion to power the "in vacuum" accessory fixtures. The unit shall provide a speed range adequate for the intended purpose. The accessory drive shall provide a self-contained power package to be used in conjunction with the headstock and tailstock assembly, tables, and the other applicable "in vacuum" accessories. Each mounting pad shall contain a four bolt tie-down pattern in conjunction with a precision pilot diameter. The motor and control system shall be feedback controlled and have a speed regulation rating within two percent of set values.

3.4.12 Closed-circuit video monitor. When specified (see 6.2), a closed circuit video monitor shall be furnished. The closed-circuit video monitor shall provide the operator a means of viewing the welding operation. This system shall consist of a camera, cathode ray tube, electrical connections, and controls. The controls shall allow the operator to focus and fine tune the picture in accordance with the manufacture's standard commercial practice. The location of the monitor shall provide the operator easy viewing. When specified (see 6.2), a video recording system shall be furnished.

3.4.13 Wire feed. When specified (see 6.2), a wire feed system shall be furnished. A wire feed system shall automatically feed wire to allow surface overlay for surface modification (hardfacing) or surface buildup (to restore worn surface areas or edges). This system shall meet or exceed the manufactures standard commercial equipment in all respects.

3.4.14 Pneumatic system. The pneumatic system shall operate on a pressure of 70 to 90 pounds per square inch gage (psig) or less and a volume of 4 cubic feet per minute (CFM) or greater. The pneumatic system shall conform to the requirements of ANSI B93.114M.

3.4.15 Coolant system. The coolant system shall assure that all components of the machine shall be capable of continuous operation at rated capacity without overheating. Enclosures shall be ventilated as required.

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3.4.15.1 Forced air cooling systems. Forced air cooling systems shall operate satisfactorily in an ambient air temperature not greater than 95 degrees Fahrenheit and shall include replaceable air filters.

3.4.15.2 Water cooling systems. Water cooling systems shall operate satisfactorily with input water temperatures up to 95 degrees Fahrenheit and a water pressure of 40 pounds per square inch gage (psig) or less. Cooling water systems shall be designed to allow smooth water flow to prevent air pockets, and to facilitate drainage. Cooling water systems shall be of corrosion resistant or non-ferrous components. Solenoid valves shall be connected by unions and shall allow water flow only as required during machine or component operation. The nameplate, or data plate, at the input water connection shall cite the water pressure, temperature, and flow requirements.

3.4.16 Electrical system. The electrical system shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), the machine shall be designed for operation on 3-phase, 60-Hertz (Hz), dual 230/460 voltage. The machine shall be initially wired as specified (see 6.2).

3.4.16.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), each motor shall have a dripproof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.17 Hour meter. When specified (see 6.2), the welding system shall be equipped with an hour meter. The hour meter shall display the accumulated time of electron beam gun operation. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the welding system.

3.5. Alignment tolerances. The welding system shall meet the alignment tolerances specified in 3.4.5.3.1, 3.4.9.1, 3.4.9.2, 3.4.9.3, 3.4.9.4, 3.4.9.5, and table I.

3.6 Performance. The system shall be capable of performing electron beam welding in a vacuum environment. The machine shall meet or exceed performance requirements of 3.4 and accuracies of table I.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally furnished with the manufacturer's commercial welding system shall be furnished, unless otherwise specified (see 6.2). All supplies and tools necessary to operate the welding machine shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

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3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2).

3.10 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in bold face on a contrasting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate. A corrosion resistant lubrication plate shall be securely attached to each welding system. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Points of lubricant application
Type of lubricant
Servicing interval
Military or Federal specification for each lubricant (if applicable)
Viscosity

3.10.2 Nameplate. A corrosion resistant nameplate shall be securely attached to each welding system. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Workmanship. Workmanship of the welding system and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by the specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall be come a part of the contractor's overall inspections system or quality program. The absence of any inspection requirements in the specification shall not

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relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The welding system shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), the quality conformance inspection shall consist of the examination in 4.7 and the test in 4.8. The machine shall pass the examination, all tests in 4.8, and the inspection to be accepted.

4.6 Acceptance test. When required (see 6.2) an acceptance test shall be performed on the welding system to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the welding system is installed at its final location. The machine shall pass an examination and all tests to be accepted.

4.7 Examination. The welding system shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 40° to 120° Fahrenheit (F) and 5 percent to 95 percent relative humidity.

4.8.2 Operational test. The welding system shall be operated under computer numerical control (CNC) for a period of 4 hours. The CNC shall cycle each axis through its entire range, addressing all speeds, feeds, direction of

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rotation, and standard control functions specified herein. Proper operation of all machine functions, controls, motors, adjustment mechanisms, and accessories shall be verified during the test period.

4.8.3 Vacuum system test.

4.8.3.1 Vacuum pump down test. The pump down performance shall be tested with the maximum workload simulated or by means of a real leak. The vacuum system shall pump the otherwise clean, dry, empty, and outgassed chamber backfilled with air, and ambient atmosphere pressure to a pressure of 1 x 10^{-4} mbar and within the time limit as specified (see 3.4.2).

4.8.3.2 Vacuum system leak test. Vacuum chamber leakage shall be tested to be in compliance with 3.4.1 with the chamber secured and the pressure at 1 x 10^{-4} mbar. No in-leakage shall be detectable at a density of 2 x 10^{-8} standard cc per second of helium.

4.8.3.3 Vacuum chamber pressure rise test. The vacuum chamber pressure rise shall be tested to be in compliance with 3.4.1. The clean, dry, empty, and outgassed vacuum chamber shall be pumped to the minimum pressure obtainable, or pumped for sixteen hours whichever requires the lesser time. The chamber shall then be valved off from the pumping system and the pressure rise monitored. After the pressure rises to 1.3 X 10^{-3} mbar, the test shall continue for four hours. The hourly rate of pressure rise shall be computed on the basis of the pressure rise data obtained in this four hour interval. The average rate of rise shall be no more than 7 X 10^{-3} mbar per hour for 10,000 liter and greater chambers, 13 X 10^{-3} mbar per hour for 1000 to 10,000 liter size chambers, and 20 X 10^{-3} mbar per hour on 1000 liter and less size chambers.

4.8.3.4 Vacuum chamber and chamber door surface finish test. The surface finish of the chamber and chamber door shall be examined to ensure a surface finish of not greater than 32 R.A. The surface finish shall be tested in accordance with ANSI B46.1.

4.8.4 Interlocks and safety devices test. The system shall be tested to be in compliance with 3.2.5.1 and 3.2.5.2.

4.8.5 Test of workpiece/gun manipulators. The manipulators shall be checked for each powered motion at 5%, 25%, 50%, 75%, and 100% of maximum movement rate with maximum load for a period of two hours, and shall not exceed the tolerance listed below:

Speed	Tolerance
5% of maximum speed	5% of set value
25% of maximum speed	2% of set value
50% of maximum speed	2% of set value
75% of maximum speed	2% of set value
100% of maximum speed	2% of set value

Workpiece or gun mechanism shall not deviate from absolute alignment of travel more than .0015 inch per foot.

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4.8.6 Performance test. The welding capability of the machine shall be tested for conformance to the requirements of 3.6 and 3.6.1. Reproducibility and reliability of the machine shall be tested over a 24 hour period. Sample welds shall be tested to be in conformance with the quality section of AMS 2681. The vacuum chamber size and the table travel shall be compatible with the material size.

4.8.6.1 Stainless steel welding test. The capability of various size machines to weld two pieces of type 302 stainless steel, each measuring three inches wide by five inches long shall be demonstrated as defined in table II. The butt weld shall join the thicknesses in the five inch dimension. The surfaces to be joined shall be cleaned prior to welding. The cross-sectional analysis of the welds performed shall show proper fusion, full penetration, and an absence of cracks and microshrinkage in accordance with AMS 2681.

TABLE II. Stainless steel welding test for Class I and Class II machines.

Size	Beam Power	Thickness of Test Pieces
1	3.0 kW	0.5 inch
2	6.0 kW	1 inch
3	7.5 kW	1 inch
4	15 kW	2 inch
5	30 kW	2 inch
6	42 kW	2 inch
7	60 kW	3 inch

4.8.6.2 Aluminum welding test. The purpose of this test shall be to produce a full penetration weld in aluminum without arcing. These bead-on-plate welds, 18 inches long, shall be made using type 5052 aluminum test pieces measuring 3 inches wide by 18 inches long as defined in table III. Immediately prior to beginning the welding test, the aluminum surfaces to be welded shall be cleaned. Any inability of any of these bead-on-plate demonstration runs to exhibit full penetration over the full length of the weld, except for no more than 1/2 inch from the start shall be cause for rejection. No arcing shall be observed during these weld demonstration tests. The weld shall conform to AMS 2681.

TABLE III. Aluminum welding test for Class I and Class II machines.

Size	Beam Power	Thickness of Test Pieces
1	3.0 kW	0.5 inch
2	6.0 kW	1 inch
3	7.5 kW	1 inch
4	15 kW	2 inch
5	30 kW	2 inch
6	42 kW	2 inch
7	60 kW	3 inch

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4.8.6.3 Dissimilar metals welding test. The capability of the welding machine to weld dissimilar metals shall be demonstrated by butt-welding a 3 inch long, 1 inch wide, 1/4 inch thick piece of copper to a piece of type 302 stainless steel of the same dimensions. The weld shall be in the 3 inch dimension. Immediately prior to beginning the welding test, the surfaces to be welded shall be cleaned. The weld shall conform to the quality requirements of AMS 2681.

4.8.6.4 Welding test for materials of varying thickness. A photomicrograph shall be used to check the weld produced by joining a clean foil strip .002 inches thick to a type 302 stainless steel block 2 inches thick. The weld shall conform to the quality requirements of AMS 2681. Failure to produce a clean sound weld shall be cause for rejection.

4.8.6.5 Additional welding test. Welds shall be made connecting specimens of like material and thickness. Butt joints shall be made in sheets of each of the specimens of materials listed in table IV. A sufficient number of welds of each material and thickness shall be made to adequately perform the tests. Immediately prior to beginning the welding test, the surfaces to be welded shall be cleaned. The weld sections (except those prepared for the tensile strength tests) shall be suitably polished and etched, where necessary, and visually examined for fusion and the presence of injurious defects such as cracks, inclusions and porosity. Test data shall include radiographs and photo- micrographs of the specimen welds. The minimum quantity of tests shall be as follows: Two radiographs, two photomicrographs, and two tensile strength tests for each thickness and each type of material. The tensile strength of welded specimens shall be not less than 90% of the strength of an unwelded sample of the same material in the annealed condition. Welds shall conform to AMS 2681. Failure of the machine to produce a sound clean weld, or failure to pass the tensile strength tests shall be cause for rejection.

TABLE IV. Additional welding test specimens.

Specimen Number	Material	Thickness inches
1	Aluminum alloy A-A 5052	0.010
2	Alloy steel SAE 4130 (cold rolled)	0.020
3	Aluminum alloy A-A 5052	0.090
4	Alloy steel SAE 4130 (cold rolled)	0.090
5	Magnesium alloy AZ63, AZ80 OR ZK60	0.500
6	Titanium alloy Ti AMS-4907D	0.090
7	Nickel alloy AMS-5545B	0.020

4.8.6.6. Reproducibility welding test. The purpose of this test shall be to produce three full penetration bead-on-plate welds that are identical but produced at different times from the same set of welding parameters. These bead-on-plate welds, 5 inches long, shall be made using type 5052 aluminum test pieces measuring 3 inches wide and 5 inches long as defined in table V.

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Immediately prior to beginning the welding test, the aluminum surfaces to be welded shall be cleaned. Micro-examinations shall be conducted from two sections of each sample. One section shall be at least one inch from the starting point, the other section at least one inch from the stopping point. The configuration and quality of the welds produced shall be identical. The welds shall conform to AMS 2681. Failure of the machine to reproduce an acceptable similar weld geometry (weld geometry whose penatration, weld width, bead size, and bead shape are within + 10 percent of being identical) from the same set of welding parameters shall be cause for rejection.

TABLE V. Reproducibility welding test data for Class I and Class II machines.

Size	Beam Power	Thickness of Test Pieces
1	3.0 kW	0.5 inch
2	6.0 kW	1 inch
3	7.5 kW	1 inch
4	15 kW	2 inch
5	30 kW	2 inch
6	42 kW	2 inch
7	60 kW	3 inch

4.8.6.7 Heat treat test. When specified (see 6.2), a sample block of AISI 1095 steel 3 inches long x 2 inches wide x 0.150 inches thick, having a hardness of Rockwell C 30 to 40 shall be heat treated in the welder in the longitudinal direction, the full length of the block. The pass shall be 0.300 inch wide. The hardness of the heat treated area shall be Rockwell C 58 minimum to 0.005 inches depth below the surface and not less than Rockwell C 50 at 0.025 inches depth in the center of the pass. Not withstanding any other provision of this contract, any chemical analysis and physical tests required by the specifications of the contract shall be made at the expense of the contractor, unless specified in the contract to be made in a Government laboratory.

4.8.6.8 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 3.6.1 shall be as specified and fully described (see 6.2).

4.8.7 Alignment tolerance test. The machine shall be tested for compliance with alignment tolerances in 3.4.5.3.1, 3.4.9.1, 3.4.9.2, 3.4.9.3, 3.4.9.4, 3.4.9.5,, and table I. The alignment tolerance test shall be performed prior to and subsequent to the performance test given in 4.8.6. Any machine adjustments required to comply with 3.4.5.3.1, 3.4.9.1, 3.4.9.2, 3.4.9.3, 3.4.9.4, 3.4.9.5 and table I during or after the performance test shall be cause for rejection.

4.8.8 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

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5. PACKAGING

5.1 Packing requirements. The requirement for packaging shall be in accordance with MIL-W-45562.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The electron beam welding system covered by this specification is used for those special welding operations requiring any or all of the following:

a. A complete absence of contaminates.

b. A narrow weld with deep penetration.

c. Precise control of the weld.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of the individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Specify system of units for measuring and indicating devices (see 3.2.1).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

i. Specify chamber size (see 3.4.1).

j. Chamber venting with pressurized bottled gas, if required (see 3.4.1.5).

k. Vacuum level, if different (see 3.4.2).

l. Specify vacuum pump down time (see 3.4.2).

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m. Vacuum tight column for moveable electron beam gun, if required (see 3.4.3).

n. Beam alignment deflection, when specified (see 3.4.3.1).

o. 1-inch, 8 channel punched tape input media, when specified (see 3.4.5.1.1).

p. RS-244 used in place of RS-358, when specified (see 3.4.5.1.1).

q. 5-1/4 inch diameter take-up spools, when specified (see 3.4.5.1.1).

r. Hardshell diskette input, when specified (see 3.4.5.1.2).

s. Specify peripheral or integrated type reader/writer (see 3.4.5.1.2).

t. Tape punch unit, when specified (see 3.4.5.2.1).

u. Ling printer, when specified (see 3.4.5.2.2).

v. Hardshell diskette, when specified (see 3.4.5.2.3).

w. Specify peripherical or integrated type reader/writer (see 3.4.5.2.3).

x. Feed rate override, when specified (se 3.4.5.3.4).

y. Specify inch or metric program data (see 3.4.5.3.15).

z. Specify CNC features, if different (see 3.4.5.3.16).

aa. Heat treat control, when specified (see 3.4.5.3.17).

bb. Specify display if other than CRT (see 3.4.6.4).

cc. Tracking system, when specified (see 3.4.7).

dd. Optical viewing system, when specified (see 3.4.8).

ee. Universal rotary table, when specified (see 3.4.9.1).

ff. Combination linear rotary table, when specified (see 3.4.9.2).

gg. Horizontal rotary table, when specified (see 3.4.9.3).

hh. Headstock and tailstock, when specified (see 3.4.9.4).

ii. Eccentric table, when specified (see 3.4.9.5).

jj. Run out platform, when specified (see 3.4.10).

kk. Accessory drive unit, when specified (see 3.4.11).

ll. Closed circuit video monitor, when specified (see 3.4.12).

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mm. Video recording system, when specified (see 3.4.12).

nn. Wire feed system, when specified (see 3.4.13).

oo. If machine is designed to operate on other than 3-phase, 60 Hertz (Hz), dual 230/460 volt electrical power, specify requirement (see 3.4.16).

pp. Specify initial wiring characteristics (see 3.4.16).

qq. Dripproof enclosure, if different (see 3.4.16.1).

rr. Hour meter, if specified (see 3.4.17).

ss. Additional performance features or performance features in lieu of requirements (see 3.6.1).

tt. Equipment furnished if other than standard (see 3.7).

uu. Specify and fully describe optional equipment, if required. (see 3.8).

vv. Specify repair parts and describe, if required (see 3.9).

ww. Information for lubrication plate, if other than listed (see 3.10.1).

xx. Information for nameplate, if other than listed (see 3.10.1).

yy. Inspection conditions, if different (see 4.3).

zz. First article inspection, if different (see 4.4).

aaa. Quality conformance inspection, if different (see 4.5).

bbb. Acceptance test, when required (see 4.6).

ccc. Location of acceptance test, if not at final location (see 4.6).

ddd. Heat treat test, when specified (see 4.8.6.7).

eee. Specify additional performance test or performance test in lieu of requirements (see 4.8.6.8).

6.2.1 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the data of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.2.1.1 Technical data. All technical data furnished should be written in the English language.

6.3 First Article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a first production item. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards. Inch-pound units having the same names in other countries may differ in magnitude.

6.4.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le System International d'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE (MIL-STD-961).

6.5 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.6 Cross-reference of classification. Classification changes from the previous issue of this specification are new. The previous issue did not list any classification of machine.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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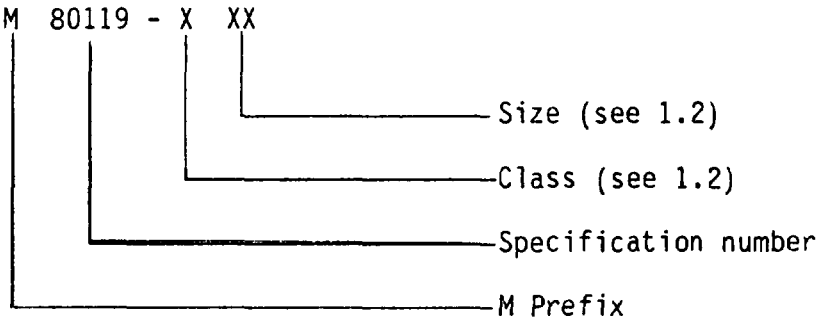
6.8 Safety and health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or identifying number (PIN). The PIN to be used for the welding system acquired to this specification are created as follows:



6.13 Subject term (keyword) listing.

Automatic welding
E. B. Welding
Electric welding
Electron beam melting
Electron beam welding
Fusion welding

Custodians:
Army - AR
Navy - SH
Air Force - 99

Preparing activity:
DLA-IP

(Project 3438-0063)

Review activities:
Army - AL, AV, ME, MI
Navy - AS
Air Force - 84

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80119B	Welding Systems, Electron Beam, Vacuum Type, CNC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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